SC Honors Geometry - Nappier
Triangle Rigidity – SSS, SAS, ASA, AAS, HL

Speak Like A Mathematician:

included angle:

included side:

hypotenuse:

leg:

congruence criterion:

The Five Triangle Congruence Criteria

SSS: if three _____ of one triangle are congruent to three sides of another, the triangles are congruent.

SAS: two sides and the _____ angle. **ASA: two angles and the _____ side.**

AAS: two angles and a _____ side. **HL: in RIGHT triangles, the _____ and one _____.**

There is no SSA and no AAA. Same angles can mean a bigger copy; SSA can leave _____ different triangles.

Example 1 – Name the Criterion

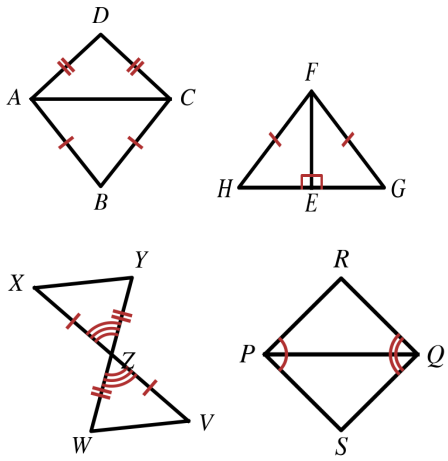
Each pair of triangles at the right is marked. Name the criterion that proves them congruent.

a) ΔABD and ΔCBD

b) ΔXYZ and ΔVWZ

c) ΔPQR and ΔPQS

d) ΔEFG and ΔEFH



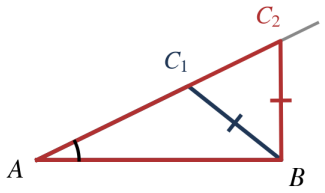
Example 2 – Why SSA and AAA Are Not Criteria

Study the two counterexamples at the right.

a) In the first figure, which three parts are congruent in both triangles? Why is that not enough?

b) In the second figure, all three angles are congruent. Why are the triangles not congruent?

c) Which criteria DO use two sides and an angle? How are they different from SSA?



same two sides and the same non-included angle — but two different triangles

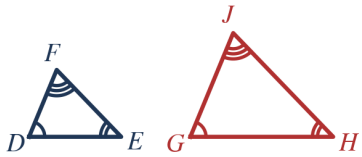
Example 3 – Find the Value of x

In the diagram, $\triangle MNO \cong \triangle PQR$.

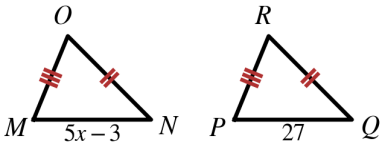
a) Which side corresponds to $\overline{MN}$?

b) Write and solve an equation for x.

c) Check your answer.



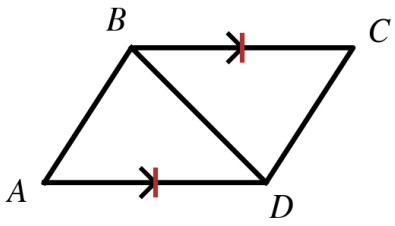
all three angles congruent — but not the same size



Example 4 – Proof Using SAS

Given: $\overline{BC} \parallel \overline{AD}$, $\overline{BC} \cong \overline{AD}$

Prove: △ABD ≅ △CDB

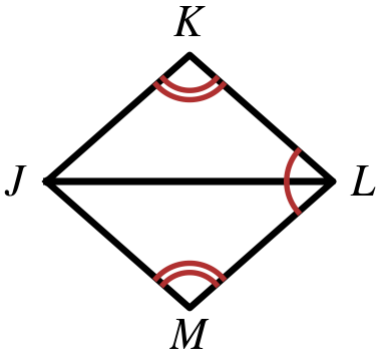


Statements	Reasons
1. $\overline{BC} \parallel \overline{AD}$	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**

Example 5 – Proof Using AAS

Given: $\overline{JL}$ bisects ∠KLM; ∠K ≅ ∠M

Prove: △JKL ≅ △JML

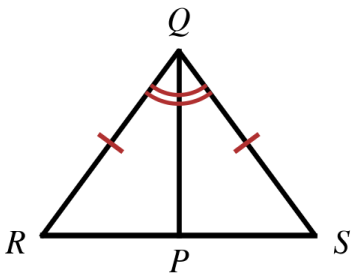


Statements	Reasons
1. $\overline{JL}$ bisects ∠KLM	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**

Example 6 – Proof Using SAS Again

Given: $\bar{Q}\bar{P}$ bisects ∠RQS; $\bar{Q}\bar{R} \cong \bar{Q}\bar{S}$

Prove: △RQP ≅ △SQP



Statements	Reasons
1. $\overline{QP}$ bisects ∠RQS	**1.** Given
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**

Example 7 – Choose a Criterion

Decide whether the given information is enough to prove the triangles congruent. If it is, name the criterion.

a) Two pairs of sides and the pair of angles between them.

b) Two pairs of angles and the pair of sides between them.

c) Two pairs of angles and a pair of sides NOT between them.

d) Two pairs of sides and a pair of angles NOT between them.

e) All three pairs of angles.